PRODUCT DATASHEET

DMXRZ-S3

Portable DMX512 & RDM Field Controller

F111 Studio Design

Document revision **DRAFT A.3** · EN v1 (specs) · 2026-10-30 (Asia/Shanghai)

Handheld **one-universe** field controller — DMX512 output, RDM address/identify, SM/UCS decoder IC write, RZ pixel drive, and Wi-Fi Art-Net bridge. Assembled, battery-powered, ready to use. **Not a lighting console.**



Figure 1. Assembled unit (handheld) — DMXRZ-S3

AT A GLANCE

110 g · one universe

- DMX512 + RDM (RS-485)
- SM / UCS IC write
- RZ pixel (WS2812 / CLK+DATA)
- Wi-Fi Art-Net node
- ~5 h runtime (Wi-Fi off)
- USB-C 5 V only

1. Cover / identity

Product	Portable DMX512 & RDM Field Controller
Model	**DMXRZ-S3**
Form factor	Battery handheld; 2.0″ color touch LCD + rotary encoder
Electrical I/O	Screw terminals: DMX/RDM **A | B | GND**; RZ **C | D | GND**; **USB-C** power / charge / firmware update
Wireless	2.4 GHz Wi-Fi station; one Art-Net universe (unicast)
Contact	Tindie listing / F111 Studio support (as published on Tindie)

HARDWARE **PCB / design V1.5**	FIRMWARE **Factory as shipped · Settings → About**	MANUFACTURER **F111 Studio Design**
DOC REVISION **DRAFT A.3 · EN v1 (specs)**	DOC DATE **2026-10-30 (Asia/Shanghai)**	STATUS **v1 · specs only**

2. Scope

This document is **edition 1** of the datasheet for the **complete assembled product as sold**. It covers mechanical envelope, power and battery, interface electricals, and absolute / recommended ratings. Capability wording follows the Tindie long-description baseline (v1.1.6).

Out of scope: feature / UI how-to; packing list; ordering and price; design-file index; firmware update procedures.

3. Product overview

DMXRZ-S3 is a **110 g** handheld field controller built around an **ESP32-S3**, with a **2.0-inch 320×240** color touchscreen, rotary encoder, protected **604050** Li-ion pack, and pluggable screw terminals (no onboard XLR socket).

Capability overview: DMX512 output, RDM, SM/UCS decoder IC write, RZ pixel drive, and Wi-Fi Art-Net (**one** universe). Optional web console on the same LAN. **Not a lighting console.**

4. Block diagram



Battery: 604050 + NTC → NTC charger (CN3, 3-pin). · **On unit:** slide switch · LEDs · beeper · 2.0″ LCD + encoder.
Orientation only. Full electrical detail: EasyEDA schematic PDF on GitHub.

5. Absolute maximum ratings

Stresses beyond these limits may cause permanent damage. Functional operation is not implied at absolute maximum. See §6 for recommended operating conditions.

Parameter	Symbol / port	Absolute maximum / limit	Notes
USB-C input voltage	V_{USB}	**5 V only**	Not for 9 V / 12 V PD or higher-voltage USB Power Delivery profiles
USB-C source current capability	—	**Min 1.5 A; recommend 2 A**	Charge + run; weak 500 mA PC ports slow/stall charge
Battery charge voltage	V_{BAT_CHG}	**4.2 V nominal charge; cell limit 4.25 V**	Protected pack; NTC-monitored charger
Battery charge / discharge rate	—	**≤ 1C**	Pack rating
DMX A / B bus common-mode (transceiver)	V_{CM}	**Approx. ±15 V (THVD1450 class)**	SMAJ12CA TVS + CMC; **not** rated for sustained **24 V / 48 V** or **mains** miswire
RZ C / D	—	**5 V logic only**	**Never** put strip power on C or D; strip must be self-powered and share GND with the unit
Ambient (cell discharge / charge window)	T_A	**0 to 50 °C**	604050 cell rating used as product window for operate/charge

> **Misuse warnings**
>
> - Do not wire the DMX line to constant-voltage LED supplies (24 V / 48 V) or mains.
> - Do not use USB-C as a high-voltage PD sink.
> - Do not connect strip V+ to RZ terminals C or D.

6. Recommended operating conditions

Parameter	Value
Operating / discharge temperature	**0 to 50 °C** (604050 cell rating)
Charging temperature	**0 to 50 °C** (604050 cell rating); charger reduces current / stops outside its NTC window
Storage temperature / humidity	Follow the **604050 cell supplier SDS / MSDS**. Store as installed equipment with Li-ion cell.
Ingress protection	**No IP rating.** Indoor / sheltered portable field and bench use.
USB-C supply	**5 V**; minimum **1.5 A**; recommended **5 V / 2 A** charger rated for 5 V output
Typical battery runtime	About **5 hours**, typical / normal use, **Wi-Fi off**. Runtime shortens with Wi-Fi + Art-Net, bright UI, and continuous high activity.
Use environment	Portable handheld; not a permanently installed outdoor fixture

7. Mechanical

Outer dimensions	**98 × 60 × 20 mm** (nominal outer envelope)
Mass	**110 g** including battery
Enclosure	**PA12** nylon (top / bottom); process **MJF** (Multi Jet Fusion)
Display opening	2.0″ LCD module (component mech DWG under LCD datasheets)
Mounting	Pocket / hand / lanyard; no rack ears
I/O	No XLR chassis socket; pluggable **3-pin screw terminals**
Outline sources	Enclosure STL: GitHub `mechanical/`; LCD module mech: `hardware/datasheets/`

> Tolerance / official 2D outline: **TBD** (nominal envelope above; see STL until a controlled 2D drawing is released).

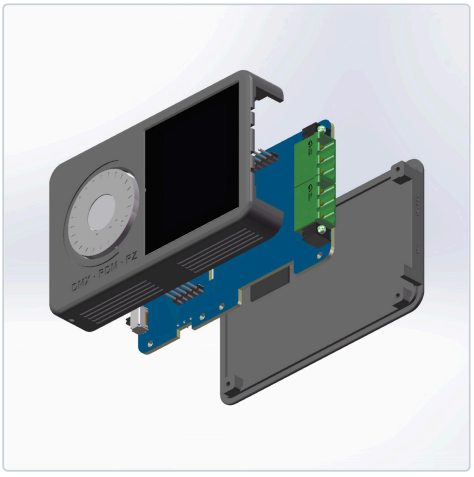


Figure 2. Exploded assembly — front shell, PCB (green terminals + USB-C), rear shell (PA12 · MJF)

8. Power & battery

Chemistry / form	Protected **1-cell** polymer Li-ion pouch
Size code	**604050** (~6 × 40 × 52 mm)
Capacity	**1500 mAh**
Nominal / charge voltage	**3.7 V** nom / **4.2 V** charge (cell limit **4.25 V**)
Energy	**5.55 Wh**
Max charge / discharge	**1C**
Transport test	**UN38.3** (report **TCT260811B001**)
Temperature sense	Pack **NTC**; charger monitors NTC, derates / stops outside window
Protection (pack)	Over-charge / over-discharge / over-current / short-circuit (as provided by protected pack)
User switch	Slide switch connects battery; with switch **off**, unit can still run from USB-C
Run while charge	Yes — USB-C powers the unit while charging
Charge time	About **2.5 h** empty→full at **1 A** from a **5 V / 2 A** source; slower (~3.5–5 h) from a 500 mA computer port; charge may stall if the unit is on and the port is weak
UI	Battery icon shows USB power, charge state, and level
Typical runtime	About **5 h** (§6)

SDS / MSDS: available on request (see §14).

9. Battery connector

Board reference	**CN3**
Connector P/N (board socket)	**HCZZ0001-3**
Pitch / poles	**1.25 mm**, **3-pin**
Signal names	**GND**, **NTC**, **VBAT**
Replacement pack	Matching **604050** protected polymer pack with **3-wire NTC** (same capacity / charge limits as §8)

> **Pin order (physical left→right on silkscreen): TBD** — confirm against the production board silkscreen / photo (not yet locked).

Safety: Replace only with a matching **3-wire NTC** protected pack. Observe polarity (wrong VBAT/GND can damage charger and pack). Power off before service; follow Li-ion handling practices.

10. Interfaces — USB-C

Power	**5 V** on USB-C
Charge	NTC-supervised charging
Firmware update	USB firmware load supported

Minimum source: **5 V / 1.5 A**. Recommended: **5 V / 2 A**.

11. Interfaces — DMX/RDM (RS-485)

11.1 Terminals & XLR map

Terminal	Signal	3-pin XLR pin
A	Data +	Pin **3**
B	Data −	Pin **2**
GND	Common / shield	Pin **1**

Polarity follows **ANSI E1.11**: A = Data+ (XLR3), B = Data− (XLR2), GND = XLR1. Some RS-485 vendors reverse A/B labels — if a fixture does not respond, check polarity first.

The unit has **no XLR socket**. Box includes a default **female–female** 3-pin XLR adapter and **2×** 3-pin pluggable terminals (cut mid-cable and land wires if needed).



Figure 3. Terminal face — RZ Code (C | D | GND) and RS-485 / DMX-RDM (A | B | GND)

11.2 Electrical

Universe	**One** (512 channels)
Termination	**133 Ω** between A and B
Isolation	**Non-isolated**; port GND = system GND
Protection stack	SMAJ12CA bidirectional TVS per data line → common-mode choke → TI **THVD1450** (transceiver ESD/EFT ratings per device datasheet; bus CM ~±15 V)
Sustained miswire	**Not** rated for 24 V / 48 V / mains

11.3 Ground loops

On battery power the unit floats — ground-loop risk is low. When USB connects to a computer or earthed charger, unit GND joins that earth through the DMX line. For long-term rig ties, prefer battery power or an **isolated** DMX splitter.

11.4 RDM and the bus

While the **RDM** page is open, RS-485 is used for discovery. After **Home** or **COLORS**, Art-Net and COLORS can drive the port again. On-device: factory RDM **on** (no separate RDM switch). Web Settings: RDM checkbox can turn RDM **off**.

12. Interfaces — RZ pixel port

Terminal	WS2812 mode	CLK+DATA mode
C	not used	**CLK**
D	**DATA**	**DATA**
GND	Common with strip ground	Common with strip ground

Logic level	**5 V** (buffered **SN74AHCT1G125**, **33 Ω** series)
ESD	Present on lines (see schematic for parts)
Power to strip	**Does not power the strip**
Wiring rule	Strip on its **own** supply; tie supply GND to unit GND; **never** put strip V+ on C or D
Factory	Wire mode **CLK+DATA**; RZ output follows output target; pixel count factory **128**

13. Wireless & Art-Net

Radio	**2.4 GHz** Wi-Fi, **station** mode (join site AP or phone/laptop hotspot)
Art-Net	**One** universe; Net **0**, Sub-Net **0**
Universe map	Unit **1–4** ↔ Art-Net **0–3**
Addressing	**Unicast required** to the unit IP (broadcast often dropped by APs)
Node name	`DMXRZ-S3` (long name `DMXRZ-S3 : Driver NO1`; part after colon can be renamed)
Hold behaviour	COLORS not Off **holds** Art-Net; when stream stops, **last frame** remains on output
RDM page	Leave RDM before Art-Net should drive the port
Tested with	**QLC+** (Art-Net → DMX fixture; Art-Net → pixel strip)
Web console	`http://[unit IP]/` — same LAN / hotspot required
Factory reset	Clears saved Wi-Fi; hardware reset button = reboot only

Universe on controller	Art-Net universe (Net 0, Sub 0)
1	0
2	1
3	2
4	3

14. Compliance & battery transport (summary)

Complete unit CE / FCC	**Not CE/FCC certified yet** (small-batch maker product). Uses an official **ESP32-S3** module that already carries RF-related certification
Enclosure material (current)	**PA12**; process **MJF** (Multi Jet Fusion). Dimensional accuracy: within **±0.3 mm** or **0.4%**; color: black / dark gray. HDT **175 °C**; elongation at break **15%**; tensile **48 MPa**; flexural **70 MPa**; impact **3.5 kJ/m²**. Formal RoHS / WEEE (etc.): **after injection tooling**.
Cell transport	Li-ion cell **UN38.3** (report **TCT260811B001**); equipment with installed Li-ion battery **5.55 Wh**. SDS / MSDS available on request.

15. Revision history

Doc rev	Date (Asia/Shanghai)	Notes
v1 (specs)	2026-10-30	Initial edition — ratings, mechanical, power, and interfaces only

End of DRAFT A.3 · EN v1 (specs). CN3 silkscreen pin order still TBD.